

Billing & Subscription page — change spec

Route: /billing · Owner: Nadeem Mohammed · Supersedes v1.0 and Addendum A · 29 July 2026

STATUS

This is the consolidated build document. It replaces spec v1.0 (28 July) and Addendum A (29 July); neither should be worked from any longer. Every rate here is signed off. Tickets BILL-01 through BILL-04 plus the new BILL-08 are the launch gate.

Area	v1.0	v2.0
Tier rates	Placeholders, pending sign-off	Signed off
Slider bands	Five, custom quote gate at 500/mo	Four, quote gate at 200/mo
Enterprise pricing	£3.20 published on the page	Quote only — never rendered
Add-on modules	Not covered	Full rate card with tier gating
What counts as one inspection	Not addressed	300-photo unit rule — new BILL-08
Top-up worked example	Two arithmetic errors	Corrected matrix, all tiers
Evidence retention	Not addressed	24-month default, extension sold as a module

1. Context — why this ships before outbound

This page becomes the landing point for cold outbound traffic. Prospects arrive from an email having spoken to nobody, and decide whether to put a card in within about ninety seconds. Every ambiguity costs a conversion.

The four defects identified in v1.0 stand:

- **Plan state contradicts itself.** Header reads "Active Plan: Growth", slider panel reads "Your Tier: Starter", credits card reads "0 Credits", top-up section reads "Used this month: 30/30". Four components, three answers.
- **The volume ladder is broken.** All three top-up packs priced at £5.50, with a "BEST VALUE" badge on the largest. It is the same value in a larger quantity.
- **Unit price is invisible.** The panel shows £49.00 but never £4.90 per inspection — the only number a buyer can benchmark against a competitor.
- **Billing period defaults to Monthly.** The annual option is discounted and better for cash collection, and it is the non-default.

A fifth defect, found since. The page bills per inspection with no upper bound on inspection size. Verified cost to serve scales almost linearly with photo count: a 250-photo inspection costs approximately £1.05 to deliver, a 600-photo inspection £2.30, and an 800-photo inspection £3.19. At the Enterprise reference rate of £3.20 an unusually photo-heavy customer is served at or below cost. BILL-08 closes this.

2. BILL-01 — one source of truth for plan state

P0 · BLOCKER · UNCHANGED FROM V1.0

Four components each derive plan state independently. Replace with a single store object. Nothing on the page computes plan state locally.

```
billingState = {
  currentPlan: {
    tier: 'growth', // what they are subscribed to NOW
    billingPeriod: 'annual', // starter | growth | professional | enterprise
    includedPerMonth: 30, // monthly | annual
    usedThisPeriod: 24, // UNITS, not raw inspections – see BILL-08
    topUpCredits: 0, // purchased packs, roll over
    unitsUnderMgmt: 1200, // drives Tenant Portal band
    renewsOn: '2026-08-23',
    status: 'active' // none | trialing | active | past_due | canceled
  },
  selectedPlan: { // what the SLIDER is previewing
    tier: 'growth', inspections: 30, billingPeriod: 'annual', modules: []
  }
}
```

Component	Reads from	Required copy
Header card 1	currentPlan	"Current plan" — tier name + billing period. If status is none, "No active plan".
Header card 2	currentPlan	"Units left" — (included – used) + topUpCredits of included + topUpCredits. Never a bare "0".
Header card 3	currentPlan	"Renews" or "Ends" depending on status.
Slider panel	selectedPlan	Label reads "Selected plan", never "Your tier".
Module cards	selectedPlan	Locked state resolves from selectedPlan.tier for preview.
Top-up section	currentPlan	Unit rate resolves from currentPlan.tier, never selectedPlan.

Rule for the whole page: anything above the slider describes what the customer has. Anything inside the slider panel describes what they are considering. Those two things must never share a label.

3. BILL-02 — slider bands, labelling, live unit price

P0 · BLOCKER · RATES NOW FINAL

Tier	Inspections / month	Annual equivalent	Rate	Annual value
Starter	10 – 29	120 – 348	£4.90	£588 – £1,705
Growth	30 – 74	360 – 888	£4.30	£1,548 – £3,818
Professional	75 – 199	900 – 2,388	£3.70	£3,330 – £8,836
Enterprise	200 +	2,400 +	By quote	£18,000 – £45,000 typical

```
TIERS = [
  { id: 'starter', min: 10, max: 29, rate: 4.90, quoteOnly: false },
```

```

    { id:'growth',      min:30,  max:74,  rate:4.30, quoteOnly:false },
    { id:'professional', min:75,  max:199, rate:3.70, quoteOnly:false },
    { id:'enterprise',  min:200, max:null, rate:null, quoteOnly:true  }
  ]

  TOPUP_MULTIPLIER = 1.10
  ANNUAL_MULTIPLIER = 0.80
  QUOTE_GATE       = 200
  PHOTOS_PER_UNIT  = 300      // see BILL-08

```

- **Add the Starter tick.** The slider starts at 10 but the first labelled tick is 30. A buyer at the default position cannot see which tier they are on. Label ticks at 10, 30, 75 and 200.
- **Always show the unit price.** Directly beneath the selected-plan line render: £{rate} per inspection · AI analysis included. This is the only figure a prospect can benchmark against a competitor and it is currently absent.
- **No hardcoded prices in components.** Every rate resolves from the config above.

4. BILL-03 — default the billing period to Annual

P0 · BLOCKER · UNCHANGED FROM V1.0

Flip the toggle. Annual becomes the pre-selected state on page load for all new subscriptions. When Annual is active the price block shows three lines, in order: annual total in largest type, monthly equivalent smaller and muted, then the saving in teal.

Selecting Monthly shows the inverse — the monthly charge, then "Switch to annual and save £387". Do not hide the saving when the customer is on the losing option; that is the moment it is most useful.

Why this is a P0 and not a nicety. A customer on £129 a month is twelve separate churn decisions and twelve separate card-failure risks. The same customer on annual is the full amount in the account on day one and no churn decision for a year. It also collapses twelve £0.20 fixed Stripe fees into one. At forty customers this is the difference between drip-feeding and having working capital.

5. BILL-04 — top-up packs, real volume ladder

P0 · BLOCKER · SUPERSEDES THE V1.0 WORKED EXAMPLE

State the rounding rule first, because it determines every figure on the cards.

```
packDiscount = { 20: 1.00, 50: 0.95, 100: 0.91 }

topUpRate    = tierRate × TOPUP_MULTIPLIER
packTotal    = round( topUpRate × packDiscount[size] × size , 0 ) // nearest £1
displayRate  = round( packTotal / packSize , 2 ) // derive from the total
```

The displayed unit rate is derived from the rounded total, not computed independently. Computing both from the raw rate produces cards where the stated rate does not multiply back to the stated price.

Tier	Base rate	Top-up base	20 pack	50 pack	100 pack
Starter	£4.90	£5.39	£108 (£5.40)	£256 (£5.12)	£490 (£4.90)
Growth	£4.30	£4.73	£95 (£4.75)	£225 (£4.50)	£430 (£4.30)
Professional	£3.70	£4.07	£81 (£4.05)	£193 (£3.86)	£370 (£3.70)
Enterprise	By quote	Contract ×1.10	Overage billed against the contract, not on the page		

Design property worth protecting: the 100-pack lands exactly on the customer's base tier rate at every tier and never below it. A customer can buy back to their contracted rate but never beneath it, so top-ups can never cannibalise a tier upgrade.

Required on each card: pack size, total price and unit rate, all three, always. On the largest pack only, a badge reading "Lowest rate" rather than "Best value" — it states a fact rather than making a claim. Beneath it: "Save £45 against buying five 20-packs." Replace "Growth tier pricing applies" with "Priced at your Growth rate".

6. BILL-08 — what counts as one inspection

P0 · BLOCKER · NEW

The page currently bills per inspection with no bound on inspection size, while cost to serve scales almost linearly with photo count. This is the largest unpriced exposure on the page.

Photos in the inspection	Verified cost to serve	Revenue at £3.20	Contribution
250	£1.05	£3.20	£2.15
300	£1.14	£3.20	£2.06
600	£2.30	£3.20	£0.90
800	£3.19	£3.20	£0.01

The fix is a unit definition rather than a price change. One inspection unit includes up to 300 photos. Inspections above that consume additional units.

```
PHOTOS_PER_UNIT = 300

unitsConsumed = MAX( 1, ROUNDUP( photoCount / PHOTOS_PER_UNIT , 0 ) )

// 1 - 300 photos    → 1 unit
// 301 - 600         → 2 units
// 601 - 900         → 3 units
```

6.1 Where this surfaces

- **Slider sub-label.** Beneath the inspection count: "Each inspection includes up to 300 photos."
- **Usage counter.** Header card 2 counts units, not inspections. "24 of 30 units used", not "24 of 30 inspections".
- **Mobile app warning at 280 photos.** "This inspection will use 2 of your monthly units." Warn during capture, never at billing. A surprise on the invoice is a support ticket and a churn risk; a warning in the field is a feature.
- **FAQ.** New item — see section 9.
- **Enterprise contracts.** The same rule, stated in the agreement. Do not leave it to be discovered.

300 is chosen, not rounded. It sits comfortably above the 250-photo working assumption, so the overwhelming majority of inspections consume exactly one unit and no customer feels metered. It holds contribution above 64% at the Enterprise reference rate. And it is a number an operations director can hold in their head.

7. BILL-09 — module cards and tier gating

P1 · REPLACES BILL-07 · RATES FINAL

The four feature cards currently render identically at every slider position, so moving the slider appears to buy quota and nothing else. Modules now have defined rates and minimum tiers, so this is configuration rather than a design judgement.

Module	Monthly	Annual (×0.80)	Available from
Tenant Portal	£100 – £300 by unit band	×0.80	Growth
Dispute Portal	£150	£1,440 / yr	Growth
Extended Evidence Retention	£75 – £250 by term	×0.80	Growth
IVY Tenant Maintenance Bot	£100	£960 / yr	Professional
IVY HQ Operations Bot	£50	£480 / yr	Professional
White Labelling	£250	£2,400 / yr	Professional

```

MODULES = [
  { id:'tenant_portal', minTier:'growth',
    bands:[ {maxUnits:250, price:100}, {maxUnits:1000, price:175},
            {maxUnits:2500, price:250}, {maxUnits:null, price:300} ] },
  { id:'dispute_portal', minTier:'growth',
    price:150 },
  { id:'retention_ext', minTier:'growth',
    terms:[ {months:48, price:75}, {months:72, price:150}, {months:null, price:250} ] },
  { id:'ivy_tenant', minTier:'professional', price:100 },
  { id:'ivy_hq', minTier:'professional', price:50 },
  { id:'white_label', minTier:'professional', price:250 }
]

```

Starter receives no modules. That is deliberate: the slider must visibly unlock capability, and the only way it does that is if the bottom band is inspections-only. At Starter all six cards render locked with their minimum-tier label. Moving to Growth unlocks three; Professional unlocks the rest.

Tenant Portal resolves from `currentPlan.unitsUnderMgmt`, not from inspection volume. These are different numbers and conflating them will mis-bill every customer whose inspection cadence differs from one per unit per year.

8. BILL-05 and BILL-06 — revenue and activation

8.1 Usage-triggered upgrade prompt

The single highest-revenue component on the page, and it does not currently exist. At 80% of the included allowance, render a banner directly above the slider panel.

```

if ( usedThisPeriod / includedPerMonth >= 0.8 ) {
  banner: "You've used {used} of {included} units this month.

```

```

    {nextTier} gives you {nextIncluded} for £{delta} more –
    and drops your rate to £{nextRate} per inspection."
  CTA:  "Move to {nextTier}" → pre-positions the slider at nextTier.min
}

```

Two behaviours matter. The banner pre-positions the slider rather than navigating away, so the customer watches the new price resolve in front of them. And the message leads with the rate drop, not the extra quota, because the rate drop is the argument that survives a finance conversation.

Instrument both events — banner rendered, banner converted. The ratio is the most important growth metric in the product and it is currently unmeasurable.

8.2 Empty state — the trial bridge

Cold outbound sends prospects a completed AI report of their own property. They arrive having already seen the output; the gap is that they must subscribe before producing a second one. For accounts with no plan and no prior trial, replace the primary CTA with "Run 3 inspections free" — no card required, reports theirs to keep — and a secondary "Or choose a plan" link that expands the slider panel. After the third free inspection, surface the panel with Starter pre-selected and the unit price shown. Habit first, card second.

9. FAQ — extend to eight items

- What happens if I exceed my monthly allowance? (existing)
- How does the annual discount work? (existing)
- Do unused inspections roll over?
- Can I change plan mid-month? — state prorating behaviour explicitly.
- How do I cancel? — link straight to the Stripe portal. Hiding this loses more sales than it saves.
- Can I export my inspection data if I leave? — for a system-of-record product this is a purchase objection, not an offboarding question.
- **What counts as one inspection?** — new. State the 300-photo rule plainly and give the worked example.
- **How long do you keep my photos and reports?** — new. State the 24-month default and that longer retention is available. A BTR compliance officer will ask this before they ask about price.

10. Acceptance criteria

- No component displays a plan name derived from anything other than billingState.
- Every tick on the slider is labelled with a tier name.
- Moving the slider updates band name, unit rate and total within one frame — no reload, no debounce above 100ms.
- A unit test asserts $\text{TTERS}[i].\text{rate} > \text{TTERS}[i+1].\text{rate}$ for every adjacent priced pair.
- Slider at $n \geq 200$ hides all pricing, renders the quote-gate block, and swaps the CTA to "Request a quote".
- No Enterprise reference rate appears anywhere in the built client bundle. Grep the built JavaScript as a release step.
- For every tier, pack display rates decrease as pack size increases: $20 > 50 > 100$.
- For every tier, the 100-pack display rate equals the tier base rate exactly. Never below.
- unitsConsumed returns 1 for 1–300 photos, 2 for 301–600, 3 for 601–900. Test the boundaries, not the middles.
- The mobile app raises the multi-unit warning at 280 photos, before capture completes.
- At Starter, all six module cards render locked with the correct minimum-tier label.
- Tenant Portal band resolves from unitsUnderMgmt, not from inspection volume.
- Annual module price uses the same ANNUAL_MULTIPLIER constant as inspections. One constant, not two.
- Annual is selected on first paint with no layout shift when the price resolves.
- Existing monthly subscribers see their current period pre-selected, not the default.
- Correct Stripe price ID passed for the selected period — verify both paths in test mode.
- Page renders correctly at 375px. Cold traffic will be roughly half mobile and the slider is the hardest element there.
- Slider is keyboard-operable with visible focus and announces the tier name on change.

10.1 Test matrix

Account state	Expected header	Expected CTA
No plan, no trial used	"No active plan" · "0 of 0"	Run 3 inspections free
No plan, trial used	"No active plan" · "0 of 0"	Start plan
Starter, 4 of 10 used	"Starter · Annual" · "6 of 10"	Disabled at slider = 10
Growth, 24 of 30 used	"Growth · Annual" · "6 of 30"	Upgrade banner visible
Growth, 30 of 30 + 20 credits	"Growth · Annual" · "20 of 50"	Upgrade banner visible
Growth, one 640-photo inspection	Counts as 3 units, not 1	Unchanged
Growth, past due	"Growth · Payment failed"	Update payment method → Stripe portal
Slider at 200 +	unchanged	Request a quote; all prices hidden

11. Build order

- **BILL-01** — state consolidation. Everything else depends on it.

- **BILL-02 and BILL-03** — slider config and annual default. Both are small once state is clean.
- **BILL-08** — photo unit rule. Touches billing, the slider sub-label, the usage counter and the mobile app, so start it early even though it is listed fourth.
- **BILL-04** — top-up ladder. Unblocked; rates are final.
- **BILL-09** — module cards and gating. Now configuration, not design.
- **BILL-05 and BILL-06** — revenue and activation mechanics.

BILL-01, 02, 03, 04 and 08 are the launch gate. Do not send outbound traffic to this page until all five are live — every prospect who lands on a page that contradicts itself is a prospect you do not get a second attempt at.